

Journal of Pharmaceutical Policy and Practice

ISSN: 2052-3211 (Online) Journal homepage: www.tandfonline.com/journals/jppp20

From transaction to transformation: a binational perspective on integrating community pharmacy into the digital health ecosystem of Southern European health systems

Diamantis Klimentidis, Zaheer Ud-Din Babar, Michael Petrides & Aliki Peletidi

To cite this article: Diamantis Klimentidis, Zaheer Ud-Din Babar, Michael Petrides & Aliki Peletidi (2026) From transaction to transformation: a binational perspective on integrating community pharmacy into the digital health ecosystem of Southern European health systems, Journal of Pharmaceutical Policy and Practice, 19:1, 2677777, DOI: 10.1080/20523211.2026.2677777

To link to this article: https://doi.org/10.1080/20523211.2026.2677777

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 03 Jun 2026.

Submit your article to this journal

Article views: 688

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at https://www.tandfonline.com/action/journalInformation?journalCode=jppp20

JOURNAL OF PHARMACEUTICAL POLICY AND PRACTICE
2026, VOL. 19, NO. 1, 2677777
https://doi.org/10.1080/20523211.2026.2677777

Taylor & Francis
by Informa

EDITORIAL

OPEN ACCESS Check for updates

From transaction to transformation: a binational perspective on integrating community pharmacy into the digital health ecosystem of Southern European health systems

Diamantis Klimentidis [a,b], Zaheer Ud-Din Babar [c], Michael Petrides [b,d] and Aliki Peletidi [b,d,e,f]

[a]Psychiatric Clinic 'Agia Aikaterini', Thessaloniki, Greece; [b]Department of Health Sciences, School of Life and Health Sciences, University of Nicosia, Nicosia, Cyprus; [c]College of Pharmacy, QU Health, Qatar University, Doha, Qatar; [d]Bioactive Molecules Research Center (BioMoReC), University of Nicosia, Nicosia, Cyprus; [e]Department of Pharmacy, Kingston University, Surrey, UK; [f]Department of Health Sciences, School of Life and Health Sciences, University of Nicosia, Athens, Greece

ABSTRACT

Background: Community pharmacists (CPs) in the binational Southern European countries of Greece and Cyprus (CG) are among the most accessible healthcare professionals, serving as key points of contact within their health systems; however, despite widespread e-prescription adoption, they remain constrained by limited access to comprehensive patient clinical data. This gap limits their ability to deliver coordinated, evidence-based care and exposes patients to avoidable medication-related harm. Within the evolving framework of the European Health Data Space (EHDS), integrating community pharmacy into national digital infrastructures represents a strategic opportunity for both nations to transition from transactional to transformative practice. This study examines international evidence and policy exemplars to advocate for granting CPs in CG full, role-appropriate, read-and-write access to patients' Electronic Health Records (EHRs), positioning them as integral actors in the digital health ecosystem.

Methods: A narrative review of global literature (2021–2025) was undertaken using PubMed, Google Scholar, and institutional sources. Studies and policies were selected for relevance to interoperability, patient safety, and governance. High-impact models, including the UK's read-write EHR pilot, Denmark's Shared Medication Record, and Estonia's national e-health infrastructure, were appraised for transferability to the CG contexts.

Findings: International evidence shows that pharmacist participation in interoperable, read-write EHRs enhances medication safety, adherence, and collaboration. Conversely, fragmented or siloed digital infrastructures – such as Greece's limited e-prescription platform or the underutilised national electronic health record system in Cyprus – fall short of enabling genuine clinical integration and coordinated, patient-centred care. A phased roadmap for both countries includes legal recognition of CPs as authorised EHR users, HL7-FHIR–based interoperability, and workforce training in digital competencies. Embedding pharmacoepidemiological indicators in pilots will yield measurable improvements in safety, efficiency, and adherence.

Conclusion: For Greece and Cyprus, granting CPs interoperable EHR access is not just digital progress but structural reform, fully integrating pharmacy into Southern Europe's digital health ecosystem.

ARTICLE HISTORY
Received 1 December 2025
Accepted 16 May 2026

KEYWORDS
Pharmacy practice reform; digital competencies; workforce development; electronic health records; patient record access; European health data space

Background

Globally, community pharmacists are recognised as among the most accessible healthcare professionals, serving as a vital frontline for public health (Muscat et al., 2024). This accessibility is particularly pronounced across Europe, which maintains a high density of pharmacists compared to other world regions (International Pharmaceutical Federation – FIP, 2021). In Greek-speaking countries (Greece and Cyprus), this accessibility is a key strategic asset; Greece has the highest pharmacy density in the EU, and Cyprus ranks second (Peletidi, Petrides, et al., 2025). Yet, these professionals operate with a profound informational handicap.

CONTACT Aliki Peletidi peletidi.a@unic.ac.cy Department of Health Sciences, School of Life and Health Sciences, University of Nicosia, Nicosia, Cyprus

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Despite its promise of delivering the right information to the right professional at the right time, digital health has yet to fully translate into practice at the pharmacy counter. This creates a daily paradox where the most frequently visited healthcare professionals are the most informationally isolated, exposing patients to the risks of fragmented care and preventable medication-related harm.

The evolution of pharmacy practice has moved far beyond the simple act of dispensing. International bodies and national health strategies increasingly envision pharmacists as integrated primary care providers who deliver a range of advanced clinical services, from medication therapy management to chronic disease monitoring and preventive care (Babar et al., 2018; Hasan et al., 2019; Scahill & Babar, 2010). Concurrently, the global transformations introduced by the Fourth Industrial Revolution are redefining pharmacy practice through automation, artificial intelligence, and data-driven healthcare systems, requiring pharmacists to develop new digital competencies and adapt to rapidly evolving care models (Baines et al., 2020). However, this clinical role is fundamentally predicated on access to comprehensive, real-time patient data. A prescription, even an electronic one, offers only a fragment of a patient's clinical history, lacking the critical context of recent diagnoses, laboratory results, hospital discharge summaries, and allergies. Operating without this information is not only inefficient but inherently unsafe, contributing to the staggering global cost of medication errors, estimated at $42 billion USD annually (WHO, 2024). This global shift toward patient-centred and system-oriented pharmacy practice has been empirically documented in a longitudinal review of FIP World Congress programmes (2003–2019), which showed sustained growth in roles emphasising professional performance, public-health engagement, and interprofessional collaboration (Kusynová et al., 2022).

This Editorial argues that granting community pharmacists full, role-appropriate, read-and-write access to patients' Electronic Health Records (EHRs) is no longer a matter of professional aspiration but a strategic necessity for Greece and Cyprus. This imperative is driven by a powerful convergence of factors: an increasingly consistent international evidence base, exemplified by recent pioneering pilots in the United Kingdom, that demonstrates clear benefits in safety and efficiency (Fynn et al., 2025); the pressing need to optimise healthcare delivery for aging populations; and the legal and political momentum of the European Health Data Space (EHDS) regulation, which provides a robust framework for secure, patient-centred data exchange across the Union (European Commission, 2025; Li & Quinn, 2024). The time has come to transform the community pharmacy from a transactional endpoint into a fully integrated clinical node in the digital health ecosystem.

Evidence appraisal and approach

This Editorial is based on a narrative review of international evidence, policy documents, and academic literature published primarily between 2021 and 2025. The aim was to synthesise high-impact exemplars and recent policy developments to inform a targeted argument for the Greek and Cypriot contexts. Sources were identified through targeted searches of PubMed, Google Scholar, and institutional websites of health authorities and professional bodies. Priority was given to peer-reviewed studies evaluating real-world implementations of pharmacist EHR access, national digital health strategies, and legal analyses of the EHDS. Consistent with recent recommendations from the International Pharmaceutical Federation's Pharmacy Practice Research Special Interest Group, future policy and implementation studies should remain grounded in robust, theory-informed, practitioner-led methodologies that can translate evidence into actionable practice (Akhtar et al., 2022; Dawoud et al., 2021). The selection was guided by relevance to the core themes of interoperability, patient safety, workflow integration, and governance.

The international evidence base

The case for pharmacist EHR access is not theoretical but it is supported by a growing body of international evidence demonstrating tangible improvements in patient safety, clinical effectiveness, and system efficiency. Recent studies provide a clear blueprint for what works, what to avoid, and how to structure implementation for success (summarised in Table 1).

Table 1. Comparative analysis of international digital health integration models for pharmacy.

Model	Description	Key Outcomes & Findings	Examples
Read-Only Viewer Portals	Pharmacists view a static summary (e.g. Summary Care Record). No ability to write back.	Benefits: Provides basic context (allergies, key diagnoses). Limitations: Passive role; interventions remain invisible to GPs. High risk of data silos.	Early UK Summary Care Record; various regional Health Information Exchanges (HIEs).
Shared Read-Write EHR	Pharmacists and GPs use the same or interoperable EHR systems, with role-appropriate permissions to both read and document clinical information.	Benefits: Enables active collaboration and electronic tasking. Limitations: High implementation complexity; requires significant investment in interoperability and training; risk of documentation duplication if not linked to reimbursement. Lesson: 'Write back' capability is essential for closing the care loop.	East of England SystmOne Pilot (UK) (Fynn et al., 2025)
National Shared Medication List	A centralised, national database providing a single source of truth for a patient's prescribed and dispensed medications, accessible to all relevant clinicians.	Benefits: Reduces reconciliation errors at care transitions. Limitations: Does not provide full clinical context (e.g. lab results, diagnoses) unless integrated within a comprehensive EHR. Lesson: Must include prescriber intent/ notes to be useful.	Denmark (Shared Medication Record) (Aharaz et al., 2023); Switzerland (eMedication Plan) (Bugnon et al., 2021).
Fully Integrated National System	A comprehensive national digital infrastructure where e-prescribing, EHRs, and pharmacy systems are fully interoperable via a central data exchange layer.	Benefits: Seamless cross-border dispensing and population health analytics.. Limitations: Highest level of political, financial, and technical investment required.	Estonia (X-Road, Prescription Centre, Health Information System) (Tuula et al., 2022).

The UK read-write access pilot

Recent evidence, such as the mixed-methods evaluation of the *SystmOne* pilot in the UK (Fynn et al., 2025) confirms that passive 'read-only' access is insufficient. In this pilot, pharmacists granted write access documented over 16,000 clinical entries, transitioning from passive viewers to active collaborators. The critical finding was that the ability to 'write back' closes the loop; interventions like stopping nephrotoxic drugs or identifying non-adherence become immediately visible to the prescribing physician, replacing untracked phone calls with auditable electronic tasking.

Similar findings have been reported beyond the UK, where qualitative evaluations from England and Australia identified general practice pharmacists as key professionals bridging gaps in multidisciplinary teams, while underscoring the need for clearer role descriptions and structured training pathways (Akhtar et al., 2022).

National integration models

While comprehensive EHR access is the goal, several European nations have demonstrated the value of integrated national systems.

The Estonian model utilizes a decentralised data exchange layer that links all pharmacies and doctors (Tuula et al., 2022). Crucially, this system supports cross-border e-dispensing, serving as a functional prototype for the EHDS. For Greece's fragmented vendor ecosystem, this demonstrates that a central database is not strictly necessary if a secure exchange layer is enforced.

Denmark's Shared Medication Record (SMR) and Switzerland's shared eMedication plan are leading examples of systems that create a single, cross-setting 'source of truth' for a patient's medications (Aharaz et al., 2023; Bugnon et al., 2021). These systems address a fundamental flaw of simple e-prescribing platforms which is no other than the lack of clinical context.

A simple list of prescriptions does not explain why a medication was started, stopped, or changed (Schiff et al., 2018). The Danish and Swiss models are designed to capture this vital context, including prescriber intent and pharmacist annotations. This creates a dynamic, living medication record that is essential for complex clinical tasks like medication reconciliation after hospital discharge and deprescribing in patients with polypharmacy. Without access to diagnoses, lab values, and the prescriber's rationale, a pharmacist cannot perform these high-stakes functions safely. This directly addresses the gaps identified by healthcare professionals in Greece, who highlighted the need for better access to clinical information like allergies,

adverse drug reactions, and drug-to-drug interactions to support their decision-making at the point of care (Grammatikopoulou et al., 2024). A shared, read-write medication record embedded within the EHR is the logical solution to this critical information gap.

However, direct transferability of these models to Southern Europe is constrained by three 'local enablers' currently maturing in Greece and Cyprus:

Legal Definitions: Unlike the UK, where pharmacists have a 'duty to access,' Greek Law has historically framed pharmacists primarily as dispensers.

Reimbursement: The UK compensates for clinical services (e.g. Pharmacy First), whereas Cypriot and Greek models are still predominantly fee-for-service, though evolving via 2025/2026 agreements.

Interoperability: Northern European systems rely on HL7 FHIR standards, while Greece's ecosystem remains fragmented with low interoperability maturity.

Lessons from the frontline: usability, workflow, and risks

International experience also provides crucial cautionary tales. A recent survey of community pharmacists in Australia (Hareem et al., 2024) that was conducted from September 2022 to June 2023 and included participants mostly from Tasmania, Victoria, and Western Australia, found that while e-prescribing was beneficial, its clinical utility was often impaired by incomplete data, sluggish software, and poor user-centred design. This finding is powerfully reinforced by the UK pilot mentioned above (Fynn et al., 2025), where the single greatest barrier to engagement was the burden of 'duplicated documentation' – requiring pharmacists to enter clinical data into the shared EHR for care coordination and then re-enter similar data into their separate pharmacy management system (PMS) for reimbursement.

Additionally, a systematic review published by Li et al. (2022) in the Journal of Medical Internet Research shows that patient safety and quality improvements are linked to genuine data liquidity, not standalone viewer portals. Furthermore, increased data access brings risks that should be managed. A key concern is alert fatigue, where an excessive number of low-specificity electronic alerts desensitises doctors, causing them to ignore both irrelevant and critical warnings. Another fact is the increased documentation burden, which can consume a significant portion of a doctor's workday if systems are not designed for efficient, workflow-aligned data entry. Any implementation must therefore include robust alert governance and prioritise a 'document once, use many times' architecture to avoid these predictable failure points.

The local context: digital progress and persistent silos in Greece and Cyprus

While international evidence provides the blueprint, successful implementation requires a strategy tailored to the unique digital health landscapes of Greece and Cyprus.

Greece's 'E-prescription paradox'

Greece's national e-prescription platform is a landmark achievement in digital health, with near-universal adoption by doctors and pharmacies (Grammatikopoulou et al., 2024). This success, however, has created an 'e-prescription paradox.' The system's efficiency in streamlining the transaction of prescribing and dispensing has inadvertently reinforced an outdated model of pharmacy practice that is exclusively medication centric. For Greek pharmacists, this system is their primary digital link to the healthcare ecosystem, cementing a relationship with prescribers that is transactional rather than collaborative.

The survey of 430 Greek healthcare professionals, including 137 pharmacists, by Grammatikopoulou et al. (2024), captured this paradox in a perfect way. While respondents praised the system for preventing errors and automating processes, they strongly highlighted the absence of crucial clinical features, such as history of allergies, adverse drug reactions, and interaction alerts, needed for robust clinical decision-making. The recent launch of the national '*MyHealth*' platform presents a historic opportunity to resolve this contradiction (Tovima.com, 2025). Functioning as the country's official national EHR, MyHealth is a patient-facing system accessible via a web portal and mobile app that centralises a citizen's key medical data, including prescriptions, referrals, laboratory results, and hospital discharge summaries. The strategic imperative is not to build a parallel system for pharmacists, but to enrich the existing, universally adopted e-prescription workflow

with read-write access to the broader clinical data now being centralised in the *MyHealth* record. This would transform the system from a transactional tool into a powerful platform for integrated clinical care.

The 'Cyprus precedent': A warning against partial implementation

Cyprus has pursued a more integrated approach, implementing a nationwide EHR as a core component of its General Health System (GHS) in 2019 (Peletidi, Petrides, et al., 2025; Petrou, 2021). However, the Cypriot experience offers a critical cautionary tale. A 2022 nationwide study of 429 doctor users (Rompolas et al., 2025) revealed only 'moderate' levels of user efficiency and, most tellingly, found that less than a third of doctors felt adequately supported by the national Health Insurance Organisation.

This 'Cyprus Precedent' is a stark warning that system availability does not guarantee effective use. The root causes of physician underutilization, identified as issues of usability, training, and institutional support, will inevitably be replicated if access is simply extended to pharmacists without addressing these fundamental problems. Therefore, the call to action for Cyprus must be twofold. First, grant community pharmacists role-appropriate access to the GHS EHR. Second, and concurrently, launch a system-wide initiative to improve EHR usability, training, and support for all healthcare professionals, leveraging the lessons learned from doctor feedback to ensure a successful and impactful rollout for the pharmacy sector.

Taken together, these international experiences demonstrate that pharmacist participation in shared electronic health records can enhance medication safety, adherence support, and clinical communication. The specific implications of these findings for Greece and Cyprus are presented in Box 1.

Box 1. Pharmacoepidemiological and adherence perspectives highlighting the real-world implications of pharmacist EHR integration in Greece and Cyprus. This box links systemic barriers to frontline risks and underscores measurable outcomes for policy and practice.

Box 1. Pharmacoepidemiology and Adherence: A Frontline Perspective

- **Medication-related harm**: Preventable errors cost over USD 42 billion globally each year (WHO, 2024). In Greece and Cyprus, ageing populations and polypharmacy heighten risks.
- **Practice realities:**
  - *Greece*: e-prescription streamlines transactions but leaves pharmacists without allergy histories, adverse drug reactions, or laboratory results.
  - *Cyprus*: A national EHR exists, yet underutilisation and poor usability limit its clinical value.
- **Adherence impact**: EHR-enabled pharmacists can monitor refill gaps against laboratory data, tailor counselling post-discharge, and document outcomes visible to the whole care team (Fynn et al., 2025). This is vital in diabetes and hypertension (Peletidi, Doundoulakis, et al., 2025; Peletidi, Petrides, et al., 2025).
- **Policy recommendation**: Future pilots should measure pharmacoepidemiological outcomes-hospitalisations avoided, adverse drug reactions reduced, adherence improved-alongside usability and adoption.
- **Human factor**: Integration must include pharmacist training in data interpretation, adherence support, and interprofessional collaboration, ensuring that digital access translates into safer, more coherent patient care.

Roadmap for integration

Overcoming the barriers to implementation requires a coordinated strategy that addresses governance, technology, and the workforce. The following roadmap, as summarized in Table 2, outlines a pragmatic, phased approach for Greece and Cyprus.

Concerns about data privacy and compliance with the General Data Protection Regulation (GDPR) are often cited as barriers to expanding EHR access. However, the European legal architecture, particularly the new EHDS regulation, reframes privacy not as a barrier, but as a core design requirement (European Commission, 2025; Li & Quinn, 2024). The legal basis for processing sensitive health data for the provision of healthcare is already well-established under GDPR Article 9(2)(h), provided it is done by or under the responsibility of a professional subject to an obligation of secrecy. The EHDS establishes a concrete, EU-wide framework for lawful data governance, incorporating safeguards such as role-based access, transparent patient-facing audit logs, and the right of patients to restrict access to specific parts of their record.

Table 2. A phased implementation framework for pharmacist EHR access in Greece and Cyprus.

Phase	Timescale	Key Objectives	Specific Actions	Lead Stakeholders
Phase 1: Foundation & Piloting	0–12 Months	Establish legal and governance foundations. Test workflows and build interprofessional trust.	Legal**:** Define pharmacists as 'authorised users' Workforce: Launch 'Digital Pharmacy Passport' training. Tech: Mandate HL7 FHIR for all pharmacy software vendors.	Ministries of Health & Digital Governance, National Health Organizations (EOPYY, HIO), National Pharmaceutical Associations.
Phase 2: Collaborative Practice	12–24 Months	Roll out 'Realistic Scope' features to support safety and communication.	Scope: Enable read/write for Allergies, OTCs, and Prescriber Notes. Workflow: Implement 'electronic tasking' to replace phone calls. Pilot: Link documentation to 'High-Cost Medicine' dispensing (Greece) and Med Rec (Cyprus).	National eHealth Authorities, PMS Vendors, Academic Institutions, Professional Regulatory Bodies.
Phase 3: Future Services	24–36 Months	Expand to advanced clinical services and public health.	Scope: Add templates for Screening (Colorectal, Smoking) and Adherence Monitoring. Data: Use structured data for population health analytics. Reimbursement: Link clinical documentation to service fees.	Ministries of Health, National Health Organizations (EOPYY, HIO), Research Institutions, National Pharmaceutical Associations.

Abbreviations: EOPYY, National Organization for the Provision of Health Services (Greece); FHIR, Fast Healthcare Interoperability Resources; HIO, Health Insurance Organisation (Cyprus); HL7, Health Level Seven International: IDIKA, e-Government Center for Social Security Services (Greece); NeHA, National eHealth Authority (Cyprus); OTC, Over-the-Counter.

To address accountability concerns within the current realities of community pharmacy in Greece and Cyprus, access privileges should focus on collaborative practice and medication safety:

Full Medication History: Pharmacists must have the ability to document Over-the-Counter (OTC) medicines and supplements alongside the prescribed list to generate a 'Best Possible Medication History' (BPMH).

Allergies and Intolerances: Read-write access to the allergy field is critical to prevent anaphylaxis and severe adverse reactions often missed in transactional dispensing.

Prescriber Communication: The system must facilitate secure, auditable communication between pharmacists and doctors to clarify orders without disrupting workflow.

Out-of-Scope: To maintain professional boundaries, pharmacists should be technically restricted from entering new medical diagnoses (ICD-10 codes) or ordering diagnostics.

Once the foundational layer of medication safety is established, the digital framework should expand to support advanced services:

Adherence & Safety Monitoring: Dedicated fields for pharmacists to document adherence barriers (e.g. side effects, cost) and clinical outcomes of chronic therapy.

Public Health Screening: Integration of structured templates for recording screenings (e.g. colorectal cancer), immunizations, and smoking cessation interventions directly into the national health record.

The policy recommendation is therefore not to invent a new privacy framework, but to faithfully implement the EHDS framework, formally recognizing community pharmacists as direct-care professionals within it.

Modern EHR platforms can readily enforce 'least-privilege' access, ensuring pharmacists see only the information relevant to their clinical duties (e.g. medications, allergies, relevant lab results, diagnoses) (Shojaei et al., 2024). By building a system with transparent, patient-visible audit trails showing 'who looked, when, and why,' Greece and Cyprus can build the public and professional trust upon which the entire system depends.

The technical challenge lies in breaking down the data silos between national EHR platforms and the proprietary PMS used in community pharmacies. As the UK and Australian experiences show, forcing pharmacists to work across multiple, disconnected systems creates unsustainable workflow friction and guarantees poor adoption (Fynn et al., 2025; Hareem et al., 2024). The solution is to mandate a 'document once, use many times' architecture.

This requires a policy-driven push for interoperability, using the EHDS's mandate for common standards as a lever. National health authorities should require that all PMS vendors adopt modern interoperability standards, such as HL7 Fast Healthcare Interoperability Resources (FHIR), to enable secure, bidirectional

data exchange with the national EHR (Ross et al., 2025; Shojaei et al., 2024; Vorisek et al., 2022). Crucially, clinical documentation must be directly linked to reimbursement. A single, structured note entered by a pharmacist for a clinical service must simultaneously populate the patient's national EHR, become visible to the GP, update the shared medication list, and automatically generate a claim for payment. Eliminating workflow duplication by mandating this level of integration is the central technical requirement for success.

Providing access to a wealth of new clinical data creates a profound shift in the professional standard of care for community pharmacy. With access comes the professional and legal 'duty to access' and use that information to ensure patient safety (Ross et al., 2025). This increased accountability means that providing access without ensuring competence would be professionally irresponsible.

Therefore, a national strategy for EHR access must be paralleled by a fully funded national strategy for workforce development. This must go beyond basic technical training. National pharmaceutical associations, in collaboration with universities, must develop and implement mandatory, accredited Continuing Professional Development (CPD) programs for all practicing pharmacists (Peletidi, Petrides, et al., 2025). These programs should focus on core competencies in clinical data interpretation, health informatics, evidence-based practice, and digital privacy. Concurrently, undergraduate pharmacy curricula must be modernised to ensure that the next generation of pharmacists graduate with these skills embedded in their training. Investing in this 'human infrastructure' is as vital as the investment in the digital infrastructure itself.

Linking evaluation to European digital health policy

Embedding pharmacoepidemiological and adherence-related indicators within national pilot evaluations aligns directly with the European Health Data Space (EHDS) vision of data-driven, patient-centred healthcare (Li & Quinn, 2024). The EHDS explicitly promotes secondary use of anonymised health data for improving quality, safety, and efficiency (European Commission, 2025), providing a regulatory pathway for transforming pilot-generated metrics into policy evidence. By systematically tracking outcomes such as reductions in preventable adverse drug reactions, improved persistence with chronic therapies, and lower readmission rates, Greece and Cyprus could position themselves as regional exemplars in leveraging EHR interoperability for measurable health gains. Moreover, linking these indicators to reimbursement and workforce development would ensure that pharmacist integration is both clinically accountable and economically sustainable. In this way, EHR access becomes not only a technological reform but a policy instrument for realising the EHDS's broader objectives-safer care, empowered professionals, and equitable access across European health systems.

Evaluating impact through pharmacoepidemiological metrics

While the proposed roadmap outlines the governance, technical, and workforce prerequisites for pharmacist EHR integration, the long-term success of these pilots will ultimately depend on measurable clinical and economic outcomes. To ensure policy sustainability and stakeholder accountability, evaluation frameworks should include pharmacoepidemiological indicators that capture the direct impact of pharmacist access on patient safety, medication adherence, and system efficiency. These metrics, summarised in Table 3, provide a pragmatic foundation for assessing both immediate and downstream benefits – ranging from reductions in adverse drug reactions (Aharaz et al., 2023) and hospital readmissions to improvements in adherence and cost savings (Li et al., 2022). Embedding such indicators within pilot evaluations will generate the robust, real-world evidence needed to justify national scale-up and inform future digital health policy across Greece and Cyprus.

Limitations

This Editorial presents a policy argument based on a narrative synthesis of existing evidence and is subject to certain limitations. The selection of international exemplars may be subject to selection bias, and we have not conducted a systematic review of all available literature. The transferability of findings from one national health system to another is complex and depends on local context, funding, and political will. Furthermore, this analysis does not include a formal cost–benefit analysis, which would be a necessary step for

Table 3. Proposed pharmacoepidemiological and practice-based indicators for evaluating the clinical, safety, and economic impact of pharmacist read–write EHR access in Greece and Cyprus.

Domain	Outcome Indicator	Operational Definition / Measurement	Expected Impact
Patient Safety	Rate of preventable adverse drug reactions (ADRs)	Number of ADR-related hospitalisations per 1,000 patients dispensed medications through participating pharmacies	↓ Reduction in preventable ADRs due to enhanced medication review and allergy checks
	Medication error rate	Number of prescription discrepancies identified and resolved by pharmacists (per 1,000 prescriptions)	↓ Improved detection and resolution of prescribing/dispensing errors
Medication Adherence	Proportion of days covered (PDC)	Percentage of patients with PDC ≥ 80% for chronic medications (e.g. antihypertensives, antiepileptics)	↑ Improved adherence through pharmacist monitoring and intervention
	Therapy persistence	Duration (in months) of continuous therapy before discontinuation or gap >30 days	↑ Increased persistence through follow-up enabled by EHR access
Health System Efficiency	Duplicated prescriptions / polypharmacy alerts resolved	Number of duplicate or unnecessary prescriptions prevented per 1,000 patients	↓ Reduced polypharmacy and therapeutic duplication
	Time to information access	Average time saved per pharmacist intervention due to integrated EHR access	↓ Reduced administrative burden, improved workflow
Economic Impact	Cost savings from avoided hospitalisations	Estimated financial savings from reduced ADR-related admissions	↑ Cost offsets and improved resource allocation
Professional Practice	Pharmacist intervention documentation rate	Number of interventions recorded in EHR per 100 patient encounters	↑ Improved accountability and interprofessional communication
	Interprofessional communication index	Number of successful task exchanges or GP acknowledgments logged electronically	↑ Strengthened collaboration and continuity of care

policymakers. Finally, the perspectives of patients and other healthcare professionals (e.g. GPs) on pharmacist data access, while acknowledged, require deeper local investigation.

Conclusion

The evidence base, while still evolving, is increasingly consistent: in health systems that grant pharmacists full, read-write, interoperable EHR access, patients can receive safer and more coherent care. The clinical, technical, and legal arguments have aligned to create a unique window of opportunity for Greece and Cyprus. The UK pilot proves feasibility and value at the clinical frontline (Fynn et al., 2025) while the Nordic, Swiss and Estonian models demonstrate the stabilizing power of national integration (Aharaz et al., 2023; Bugnon et al., 2021; Tuula et al., 2022). On the other hand, Australia's experience provides a clear lesson on the importance of user-centred design (Hareem et al., 2024). Finally, the European Health Data Space provides the legal engine to make this transformation lawful, portable, and patient-centred (European Commission, 2025; Li & Quinn, 2024).

The path forward is clear. The Ministries of Health and Digital Governance in Greece and Cyprus must act decisively to formally recognise community pharmacists as integral primary care providers within the national digital health strategy. This requires granting them full, role-appropriate EHR access, mandating the technical interoperability needed to create a seamless workflow, and investing in the training required to ensure the workforce is competent and accountable. This is not merely an IT project; it is a fundamental reform of primary care. By breaking down the digital silos that have long isolated their most accessible clinicians, Greece and Cyprus can build a more resilient, efficient, and patient-centric healthcare system.

Acknowledgements

The authors would like to acknowledge that there are no additional acknowledgements for this work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Diamantis Klimentidis http://orcid.org/0000-0003-2149-3644

Zaheer Ud-Din Babar http://orcid.org/0000-0003-3664-8901
Michael Petrides http://orcid.org/0000-0003-0388-3243
Aliki Peletidi http://orcid.org/0000-0002-8703-497X

References

Aharaz, A., Kejser, C. L., Poulsen, M. W., Jeftic, S., Ulstrup-Hansen, A. I., Jørgensen, L. M., Iversen, E., Thorhauge, A. M., & Houlind, M. B. (2023). Optimization of the Danish national electronic prescribing system to improve patient safety: Development of a user-friendly prototype of the digital platform shared medication record. *Pharmacy*, *11*(2), 41. https://doi.org/10.3390/pharmacy11020041

Akhtar, N., Hasan, S. S., & Babar, Z. U. D. (2022). Evaluation of general practice pharmacists' role by key stakeholders in England and Australia. *Journal of Pharmaceutical Health Services Research*, *13*(1), 31–40. https://doi.org/10.1093/jphsr/rmac002

Babar, Z.-U.-D., Scahill, S., Nagaria, R. A., & Curley, L. E. (2018). The future of pharmacy practice research – Perspectives of academics and practitioners from Australia, NZ, United Kingdom, Canada and USA. *Research in Social and Administrative Pharmacy*, *14*(12), 1163–1171. https://doi.org/10.1016/j.sapharm.2018.01.004

Baines, D., Nørgaard, L. S., Babar, Z.-U.-D., & Rossing, C. (2020). The fourth industrial revolution: Will it change pharmacy practice? *Research in Social and Administrative Pharmacy*, *16*(9), 1279–1281. https://doi.org/10.1016/j.sapharm.2019.04.003

Bugnon, B., Geissbuhler, A., Bischoff, T., Bonnabry, P., & von Plessen, C. (2021). Improving primary care medication processes by using shared electronic medication plans in Switzerland: Lessons learned from a participatory action research study. *JMIR Formative Research*, *5*(1), e22319. https://doi.org/10.2196/22319

Dawoud, D., Chen, A. M. H., Rossing, C. V., Garcia-Cardenas, V., Law, A. V., Aslani, P., Bates, I., Babar, Z.-U.-D., & Desselle, S. (2021). Pharmacy practice research priorities during the COVID-19 pandemic: Recommendations of a panel of experts convened by FIP pharmacy practice research special interest group. *Research in Social and Administrative Pharmacy*, *17*(1), 1903–1907. https://doi.org/10.1016/j.sapharm.2020.08.020

European Commission. (2025). *European Health Data Space Regulation (EHDS)*. https://health.ec.europa.eu/ehealth-digital-health-and-care/european-health-data-space-regulation-ehds_en.

Fynn, J., Lamptey, M., Coote, G., Twigg, M. J., Newman, J., Lingard, S., Cooper, M., & Nazar, H. (2025). A mixed methods evaluation of a shared electronic health record between general practice and community pharmacy. *International Journal of Clinical Pharmacy*, https://doi.org/10.1007/s11096-025-01972-6

Grammatikopoulou, M., Lazarou, I., Giannios, G., Kakalou, C. A., Zachariadou, M., Zande, M., Karanikas, H., Thireos, E., Stavropoulos, T. G., Natsiavas, P., Nikolopoulos, S., & Kompatsiaris, I. (2024). Electronic prescription systems in Greece: A large-scale survey of healthcare professionals' perceptions. *Archives of Public Health*, *82*(1), 68. https://doi.org/10.1186/s13690-024-01304-6

Hareem, A., Stupans, I., Park, J., Soo, A., Stevens, J. E., & Wang, K. (2024). Electronic health records and e-prescribing in Australia: An exploration of technological utilisation in Australian community pharmacies. *International Journal of Medical Informatics*, *187*, 105472. https://doi.org/10.1016/j.ijmedinf.2024.105472

Hasan, S. S., Thiruchelvam, K., Kairuz, T., Abbas, N., & Babar, Z. U. D. (2019). Pharmacy practice and Its research: Evolution and definitions. *Encyclopedia of Pharmacy Practice and Clinical Pharmacy*, 1–6. https://doi.org/10.1016/B978-0-12-812735-3.00623-3

International Pharmaceutical Federation – FIP. (2021). *Community pharmacy at a glance 2021 – Regulation, scope of practice, remuneration and distribution of medicines through community pharmacies and other outlets*.

Kusynová, Z., van den Ham, H. A., Leufkens, H. G. M., & Mantel-Teeuwisse, A. K. (2022). Longitudinal study of good pharmacy practice roles covered at the annual world pharmacy congresses 2003–2019. *Journal of Pharmaceutical Policy and Practice*, *15*(1), 94. https://doi.org/10.1186/S40545-022-00482-4

Li, E., Clarke, J., Ashrafian, H., Darzi, A., & Neves, A. L. (2022). The impact of electronic health record interoperability on safety and quality of care in high-income countries. *Systematic Review. Journal of Medical Internet Research*, *24*(9), e38144. https://doi.org/10.2196/38144

Li, W., & Quinn, P. (2024). The European health data space: An expanded right to data portability? *Computer Law & Security Review*, *52*, 105913. https://doi.org/10.1016/j.clsr.2023.105913

Muscat, N. A., Sinclair, P., Zapata, T., Connolly, D., Pinto, G. S., & Kniazkov, S. (2024). Embracing pharmacists' roles in healthcare delivery. *The Lancet Regional Health – Europe*, *46*, 101088. https://doi.org/10.1016/j.lanepe.2024.101088

Peletidi, A., Doundoulakis, I., Vavoulioti, E., Petrou, C., & Petrides, M. (2025). Hypertension care: The knowledge and attitudes of the community pharmacists. *Journal of Pharmaceutical Policy and Practice*, *18*(1), https://doi.org/10.1080/20523211.2025.2544635

Peletidi, A., Petrides, M., Birlirakis, V., Klimentidis, D., Liaras, K., & Petrou, C. (2025). Advanced clinical services in community pharmacies: Training challenges and real-world applications – A focus on Greek-speaking countries. *Journal of Pharmaceutical Policy & Practice*, *18*(1), 2455066. https://doi.org/10.1080/20523211.2025.2455066

Petrou, P. (2021). The 2019 introduction of the new national healthcare system in Cyprus. *Health Policy*, *125*(3), 284–289. https://doi.org/10.1016/j.healthpol.2020.12.018

Rompolas, P., Masouras, P., Avgousti, S., & Charalambous, A. (2025). Efficiency of electronic health record users in the general health system of Cyprus. *Health Informatics Journal*, *31*(3), https://doi.org/10.1177/14604582251353418

Ross, L. A., Dumont, Z., Herbert, S. M. C., Irwin, A. N., McDonough, R., Anderson, L., Furlow, K. W., Hamper, J., Hodge, C. K., Kier, K. L., Daniel, J. M., & Pan, I. (2025). Framework and recommendations to support bidirectional information sharing between community pharmacies and other health care settings. *JACCP: Journal of the American College of Clinical Pharmacy*, *8*(5), 406–412. https://doi.org/10.1002/jac5.70027

Scahill, S. L., & Babar, Z. U. D. (2010). Community pharmacy practice in high and low income countries: Commonalities, differences and the tension of being "retailer" versus "primary health care provider". *Southern Med Review*, *3*(2), 1–2. https://pure.hud.ac.uk/en/publications/community-pharmacy-practice-in-high-and-low-income-countries-comm/.

Schiff, G., Mirica, M. M., Dhavle, A. A., Galanter, W. L., Lambert, B., & Wright, A. (2018). A prescription for enhancing electronic prescribing safety. *Health Affairs*, *37*(11), 1877–1883. https://doi.org/10.1377/hlthaff.2018.0725

Shojaei, P., Vlahu-Gjorgievska, E., & Chow, Y.-W. (2024). Security and privacy of technologies in health information systems: a systematic literature review. *Computers*, *13*(2), 41. https://doi.org/10.3390/computers13020041

Tovima.com. (2025). *Greece Launches National Electronic Health Record Platform*. Tovima.Com. https://www.tovima.com/society/greece-launches-national-electronic-health-record-platform.

Tuula, A., Sepp, K., & Volmer, D. (2022). E-solutions in Estonian community pharmacies: A literature review. *Digital Health*, *8*, 205520762211137. https://doi.org/10.1177/20552076221113731

Vorisek, C. N., Lehne, M., Klopfenstein, S. A. I., Mayer, P. J., Bartschke, A., Haese, T., & Thun, S. (2022). Fast healthcare interoperability resources (FHIR) for interoperability in health research. *Systematic Review. JMIR Medical Informatics*, *10*(7), e35724. https://doi.org/10.2196/35724

WHO. (2024). *Global burden of preventable medication-related harm in health care: A systematic review*. World Health Organization.